

Nicolas Dickenmann

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Machine Learning researcher specializing in 3D computer vision and generative models, with publications at CHI and research experience at Stanford's Gradient Spaces Lab.

EDUCATION

Stanford University Visiting Student Researcher at the Gradient Spaces Lab	Feb 2026 - Aug 2026
ETH Zurich M.Sc. in Data Science with a focus on Computer Vision and Scalable Computing B.Sc. in Electrical Engineering, Oct 2024. GPA: 5.6/6	Sep 2021 - Aug 2026
National University of Singapore Semester abroad, completed full semester workload of EECS courses. GPA: 3.74/4	Feb 2024 - Jun 2024

RESEARCH & PROJECTS

Steering Generative Models for Accessibility: EasyRead Image Generation - Built an EasyRead pictogram generation pipeline by fine-tuning Stable Diffusion with LoRA on a multi-dataset corpus and introducing an EasyRead quality metric, enabling scalable, accessibility-focused visual content generation. Accepted at ACM CHI26 (premier Human-Computer Interaction conference)	Oct 2025 - Feb 2026
3D shape assembly pipeline (Stanford's Gradient Spaces Lab) - Extended the existing multi-part shape assembly baseline (Rectified Point Flow) with a multi-stage framework, enabling successful reassembly of objects with up to 115 parts (previously below 64), achieving a 4× reduction in rotational error and a 38% decrease in translation error - Developing an agentic framework for point cloud assembly using patch embeddings and flow model reasoning.	July 2025 - present
FP64 Emulation Engine for CUDA Option Pricing - Engineered an emulated-double arithmetic solution using paired single-precision (FP32) values to bypass the 64:1 FP32-to-FP64 performance penalty on target GPU architectures achieving 48 bits of effective precision and delivering a 3.8x speedup over native double-precision. - Wrote a blog on FP64 emulation that reached the top of hackernews.	May 2024 - Oct 2024
Bachelor's Thesis - A RL Approach to Generate Network Topologies (Grade: 6/6) - Developed an architecture from scratch utilizing Monte Carlo Tree Search to generate interconnect topologies of diameter-3 for data centers. Profiled, optimized and parallelized the architecture for a 10× performance increase over the naive implementation. - Part of my work served as foundation for this paper (see acknowledgments).	May 2024 - Oct 2024

EXPERIENCE

AI-native Car Marketplace <i>Co-Founder</i> - Built an MVP enabling semantic vehicle search using embedding-based retrieval combined with structured filtering. - Secured \$80k in angel commitments, pitched to VCs, and talked to dealerships to understand their needs.	Dec 2024 – Apr 2025
Polybee <i>Computer Vision Engineer, Intern</i> - Enabled precise crop monitoring by building a point cloud completion model for occluded spinach leaves, achieving ± 0.3 cm leaf length accuracy from RGB-D inputs. - Trained a time series model to forecast spinach leaf size distribution up to 3 days ahead, achieving 90% historical accuracy within ± 0.5 cm.	Singapore Nov 2024 – Feb 2025

AWARDS

- CHI 2026** — Paper Acceptance
- Hack&Roll 2025** — Overall Winner (Singapore's largest student hackathon)
- YC AI Startup School** — Invitee (2025)
- Swiss Study Foundation** — Scholar
- Swiss National Science Competition** — Winner (2021)

SKILLS

- Languages:** Python, C++, C
- ML Areas:** Diffusion & Flow Models, Generative Models, 3D Computer Vision, Reinforcement Learning (MCTS)
- Tools and Libraries:** PyTorch, Cuda, Wandb, Bash, Nvidia Nsights, Git, Slurm